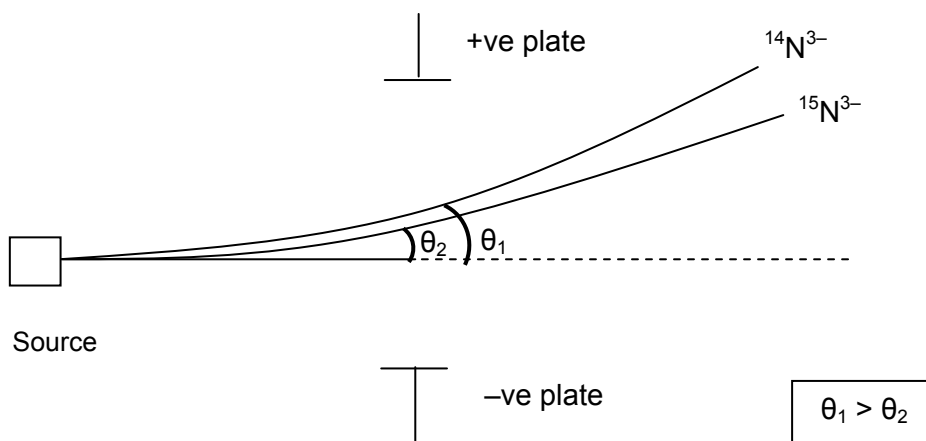


- 1 (a) (i) The mass of one atom of an isotope of an element relative to $1/12$ the mass of an atom of carbon-12.

(ii) A_r of N = $(0.9963 \times 14) + (0.0037 \times 15)$
 = **14.004**

(iii)



Both $^{14}\text{N}^{3-}$ and $^{15}\text{N}^{3-}$ have the same charge of -3 . Since angle of deflection, $\theta \propto \frac{\text{charge}}{\text{mass}}$, $^{14}\text{N}^{3-}$ is deflected to a larger angle towards the positive plate as it is lighter in mass.

- (b) (i) Element **A**

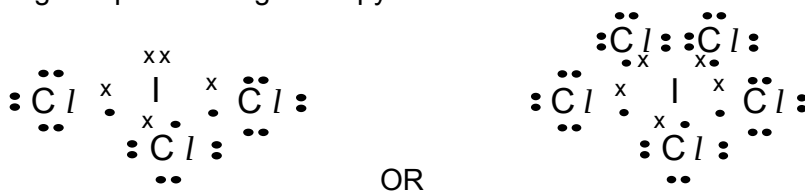
- (ii) Electronic configuration of **B**: $1s^2 2s^2 2p^4$
 Electronic configuration of **J**: $1s^2 2s^2 2p^6 3s^2 3p^4$

The electron removed from **B** is from 2p orbital, whereas the electron removed from **J** is from 3p orbital which is at a higher energy level, much lesser energy is required to remove electron from **J** than from **B** and hence a decrease in first ionisation energy.
OR

The electron removed from **B** is from 2p orbital, whereas the electron removed from **J** is from 3p orbital. Since **J** has a larger atomic radius than **B**, the valence e^- in **J** experience a smaller effective nuclear charge effect and hence less attracted towards nucleus. Much lesser energy is required to remove electron from **J** than from **B** and hence a decrease in first ionisation energy.

- (iii) Group II
 FH_2

- (iv) ICl_3 / ICl_5
 Trigonal planar / Trigonal bipyramidal



(v) The electron removed from **I** is an unpaired electron in 3p orbitals whereas the electron removed from **J** is from paired electrons in 3p orbital which experiences inter-electronic repulsion. Lesser energy is hence required to remove the electron from Y.

(c) Ammonia is more soluble in water compared to methane. Ammonia can form hydrogen bonding with water molecules whereas methane only has instantaneous dipole-induced dipole interactions which are not strong enough to overcome the hydrogen bonding in water molecules.

2 (a) (i) ΔH_1 is the sum of 1st and 2nd ionisation energy of Ba.
 ΔH_2 is the (twice) enthalpy change of neutralisation.

(ii) $\Delta H_1 = 1^{\text{st}}$ ionisation energy of Ba + 2nd ionisation energy of Ba
 $= 502 + 966$
 $= \mathbf{1468 \text{ kJ mol}^{-1}}$

(iii) $\Delta H_{\text{rxn}} = +182 + \Delta H_1 - (\Delta H_2) + (-1305) + (-796)$
 $= +182 + 1468 - (-116) + (-1305) + (-796)$
 $= \mathbf{-335 \text{ kJ mol}^{-1}}$

(iv) $Q = mc\Delta T$
 $= (100 \times 1.00)(4.18)(33.5 - 22.0)$
 $= 4807 \text{ J}$
 $\approx 4810 \text{ J (3 s.f.)}$

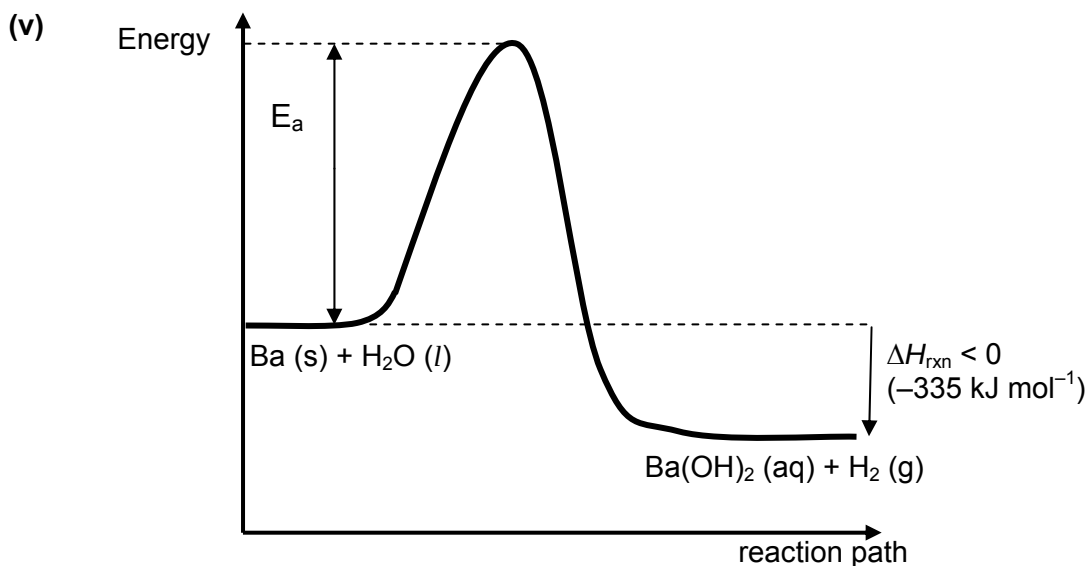
$$Q = \Delta H_{\text{rxn}} \times \eta_{\text{Ba}}$$

$$4810 = (335 \times 1000) \times \frac{m}{M}$$

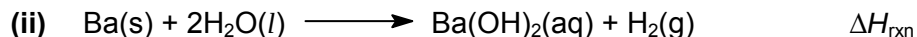
$$\text{Mass of barium added} = \frac{4810 \times 137}{335000}$$

$$= 1.966$$

$$= \mathbf{1.97 \text{ g}}$$

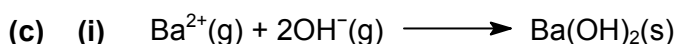


- (b) (i) Standard enthalpy change of formation is the amount of heat absorbed or evolved when one mole of a substance is formed from its constituent elements, all in their standard states at 298 K and 1 atm.



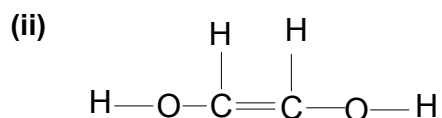
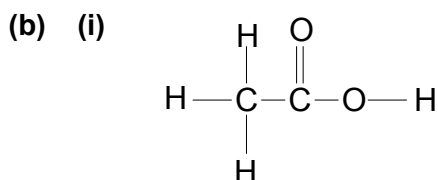
$$\begin{aligned}\Delta H_{\text{rxn}} &= \Sigma \Delta H_f (\text{products}) - \Sigma \Delta H_f (\text{reactants}) \\ &= \Delta H_f \text{Ba(OH)}_2\text{(aq)} - 2(\Delta H_f \text{H}_2\text{O(l)})\end{aligned}$$

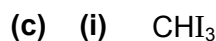
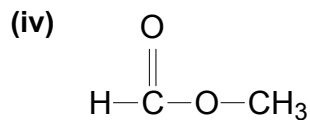
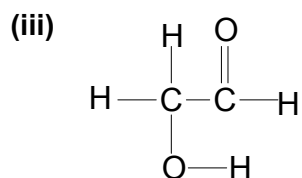
$$\begin{aligned}\Delta H_f \text{Ba(OH)}_2\text{(aq)} &= \Delta H_{\text{rxn}} + 2(\Delta H_f \text{H}_2\text{O(l)}) \\ &= -335 + 2(-285.8) \\ &= \mathbf{-906.6 \text{ kJ mol}^{-1}}\end{aligned}$$



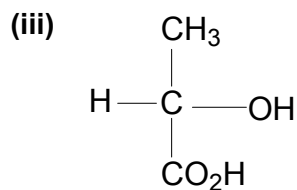
- (ii) The lattice energy of Ba(OH)_2 is less exothermic than Sr(OH)_2 . Ba^{2+} has the same charge as Sr^{2+} but a larger ionic radius, hence the electrostatic forces of attraction between Ba^{2+} and OH^{-} will be weaker than that between Sr^{2+} and OH^{-} . Since $\text{LE} \propto \frac{q^+q^-}{r^+ + r^-}$, the LE of Ba(OH)_2 is less exothermic than that of Sr(OH)_2 .

3 (a)	Compound	Description of test	Observations
	CH_2Cl_2	<u>Heat with aq NaOH.</u> (Cool) Neutralise mixture with excess dil HNO_3 . Add <u>aq AgNO₃</u> .	<u>White ppt</u> formed (AgCl)
	$\text{HOCH}_2\text{CH}_2\text{OH}$	Add anhydrous PCl_5 . or, Add sodium metal or, heat with acidified $\text{K}_2\text{Cr}_2\text{O}_7$ or heat with acidified KMnO_4	Steamy white fumes (HCl) effervescence that gives pop sound with lighted splinter(H_2) orange solution turns green purple solution turns colourless

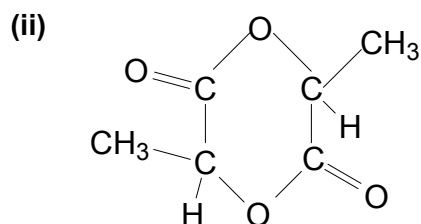




(ii) ketone, carboxylic acid



(d) (i) condensation / esterification



4 (a) Acids are proton donors.
Bases are proton acceptors

(b) (i) Weak acid.

$$[\text{H}^+] = 10^{-2.43} = 3.72 \times 10^{-3} \text{ mol dm}^{-3} < 0.1 \text{ mol dm}^{-3}$$

Lactic acid is partially dissociated.

(ii) $K_a \approx [\text{H}^+]^2 / [\text{HA}]_{\text{initial}}$
 $= (3.72 \times 10^{-3})^2 / (0.1)$
 $= 1.38 \times 10^{-4} \text{ mol dm}^{-3}$

OR

	HA	$\rightleftharpoons$	H^+	+	A^-
initial concentration / mol dm ⁻³	0.1		0		0
change in conc / mol dm ⁻³	$-10^{-2.43}$		$+10^{-2.43}$		$+10^{-2.43}$
concentration at eqm / mol dm ⁻³	$0.1 - 3.72 \times 10^{-3}$		3.72×10^{-3}		3.72×10^{-3}

$$\begin{aligned}
 K_a &= [\text{H}^+]^2 / (0.1 - 3.72 \times 10^{-3}) \\
 &= (3.72 \times 10^{-3})^2 / (0.1 - 3.72 \times 10^{-3}) \\
 &= 1.43 \times 10^{-4} \text{ mol dm}^{-3}
 \end{aligned}$$

(iii) phenolphthalein

(iv) amount of $\text{Ca}(\text{OH})_2$ reacted = $0.250 \times 22.3/1000$
= 5.575×10^{-3} mol

$$n(\text{lactic acid}) = 2 \times 5.575 \times 10^{-3}$$
$$= 0.01115 \text{ mol}$$

$$M_r \text{ of lactic acid} = 1 / 0.01115$$
$$= \mathbf{89.7}$$

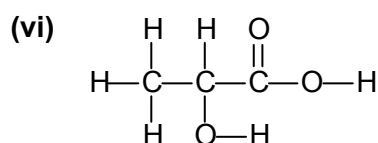
(v)

	C	H	O
Mass /g	40.0	6.7	53.3
No. of moles	$40.0/12 = 3.3$	6.7	$53.3/16 = 3.3$
Ratio	1	2	1

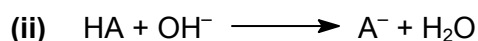
Empirical formula is CH_2O

$$n[12.0 + 2(1.0) + 16.0] = 89.7$$
$$n = 3$$

Molecular formula is $\mathbf{C_3H_6O_3}$



(c) A buffer solution is one who pH remains almost constant when small amounts of acid or alkali are added to it.



(e) At max buffer capacity, $[\text{A}^-] = [\text{HA}] = 0.1 \text{ mol dm}^{-3}$

$$\text{No. of moles of lactate ion} = 0.1 \times 500/1000 = 0.05$$

$$\text{Mass of sodium lactate needed} = 0.05 \times [(23.0 + 3(12.0) + 5(1.0) + 3(16.0))$$
$$= \mathbf{5.60 \text{ g}}$$

5 (a) (i) Substitution

(ii) A large part of TNT molecule is non-polar / hydrophobic.

(b) (i) KMnO_4 and dil. H_2SO_4 , heat

(ii) Benzoic acid has higher boiling point than methylbenzene.

Benzoic acid molecules form strong hydrogen bonds that require more energy to break compared to the weaker VdW forces between methylbenzene molecules.

- (iii) The electronegative Cl atom of 3-chlorobenzoic acid will exert an electron-withdrawing effect on the carboxylate anion.

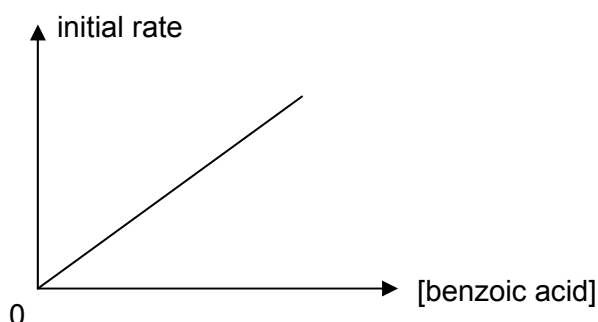
The negative charge becomes more dispersed and the ion becomes more stable.

Hence the dissociation of 3-chlorobenzoic acid to give H^+ is more favoured.

- (iv) It produces Cl^+ which will increase concentration of Cl^+ / facilitate the substitution of H in benzene / increase attraction to the electron-rich benzene hence increases the rate of formation of 3-chlorobenzoic acid.

- (c) (i) The term “order of reaction” refers to the power to which the concentration of the reactant is raised in the experimentally determined rate equation.

(ii)



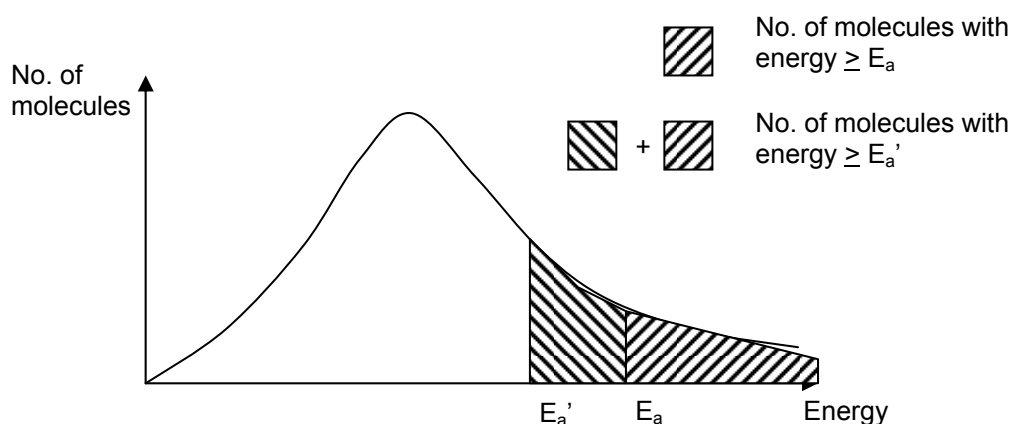
- (iii) First order with respect to chlorine.

When concentration of chlorine was halved, the gradient at each point was half that obtained in (ii) suggests that rate of reaction was halved.

- (iv) $\text{Rate} = k [\text{benzoic acid}] [Cl_2]$

- (v) $\text{mol}^{-1} \text{dm}^3 \text{s}^{-1}$ or $\text{mol}^{-1} \text{dm}^3 \text{min}^{-1}$

(d)



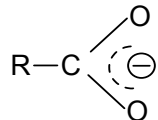
With a catalyst, activation energy of the catalysed reaction is lowered ($E_a' < E_a$) More molecules possess energy greater than or equal to E_a' .

Frequency of effective collision increases thus rate increases.

- 6 (a) The alkyl group in the alkoxide anion is electron-donating, thus intensifying the negative charge on the oxygen atom in the anion. This effect destabilises the anion, thus dissociation of the alcohol into alkoxide ion and H^+ ion is not favoured.

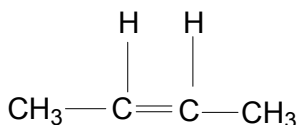
(Note: In fact, alcohols are not acidic, they are even weaker acids than water.)

The negative charge in the carboxylate anion is delocalised over the two O atoms,

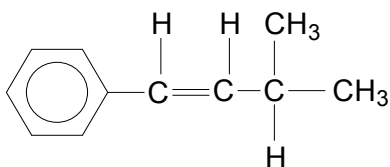


This stabilises the anion and facilitates the dissociation of the carboxylic acid into its ions. Thus, ethanoic acid is a stronger acid than ethanol.

- (b) (i)



- (ii)



(c) No. of moles of acid = $\frac{0.37}{5(12.0) + 8(1.0) + 5(16.0)}$
 $= 2.50 \times 10^{-3} \text{ mol}$

No. of moles of NaOH = $\frac{25.0}{1000}(0.20)$
 $= 5.00 \times 10^{-3} \text{ mol}$

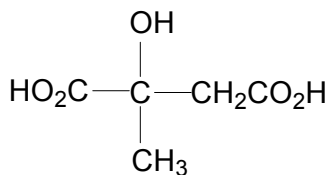
Hence, 2 mol NaOH reacts with 1 mol **T**
 $\Rightarrow$ **T** is a dibasic acid / **T** has two $-\text{CO}_2\text{H}$ groups

No reaction with acidified $\text{K}_2\text{Cr}_2\text{O}_7$ shows **T** does not have primary alcohol, secondary alcohol or aldehyde groups.

No reaction with aq Br_2 shows no alkene group in **T**.

Since 1 mol of **T** $\equiv$ 3 mol sodium, **T** has 3 $-\text{OH}$ groups.

Structure of **T** is



- (d) (i) Heat under reflux, concentrated H_2SO_4 (catalyst)

(ii) M_r of **T** = 148

Let M_r of **U** be x .

$$(148 + 2x) - 2(18.0) = 232$$

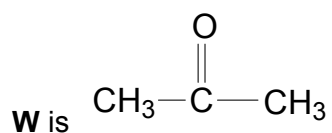
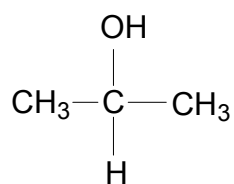
$$x = 60$$

(iii) **W** reacts with 2,4-DNPH $\Rightarrow$ it is a carbonyl compound

W has no reaction with Tollen's $\Rightarrow$ not an aldehyde

Thus **W** is a ketone
and **U** is a secondary alcohol.

And since **U** has an M_r of 60, structure of **U** is



V is

